TRACTIAN

Industrial Electric Motors

A practical and comprehensive guide on the application of electric motors in the industry

Learn about types of motors, components, common faults, and practical ways to avoid failures.



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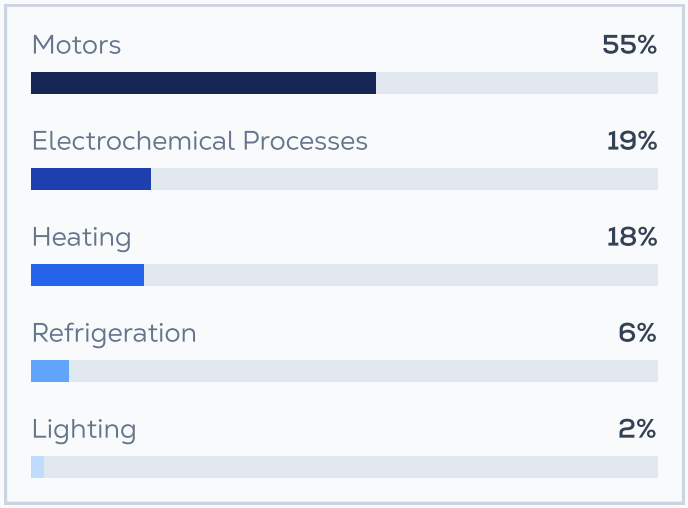
INTRODUCTION

What Is an Electric Motor?

An electric motor is a device that converts electrical energy into mechanical movement. Comprised of components such as coils, rotors, and stators, the electric motor plays a crucial role in various applications, driving machines and equipment.

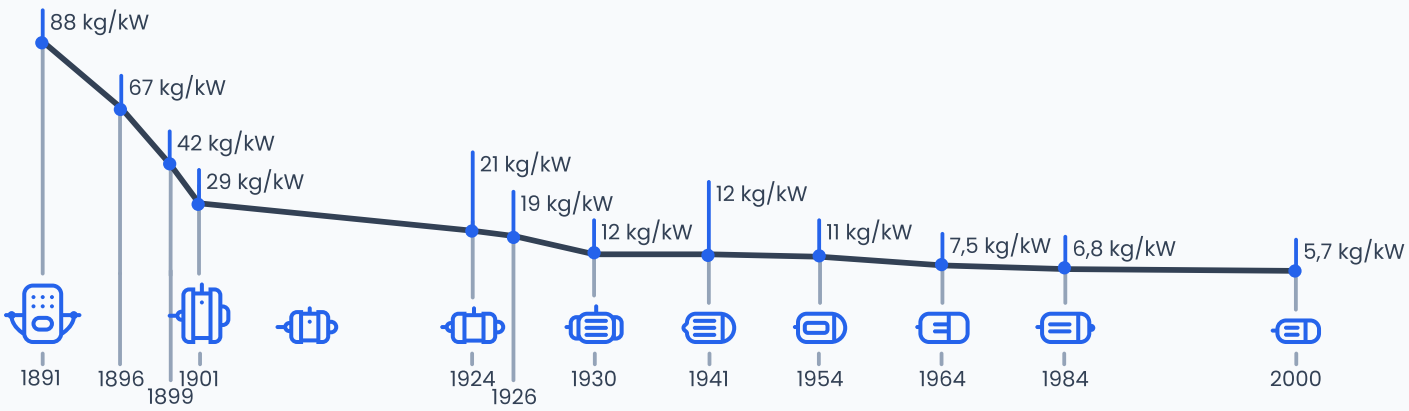
The electric motor is capable of transforming electrical energy into mechanical energy, utilizing the principle of interaction between magnetic fields.

They are also the largest energy consumers within industries and manufacturing. Check out the chart showing the use of electrical energy in the industrial sector:



The Technological Evolution of Motors

Electric motors function as the "heart" of any factory. They drive assembly lines, power conveyor belts, and provide mobility to robotic arms and joints. As they are used in almost all industrial environments, motors are becoming increasingly complex and technical, and making them perform at peak performance without compromising their integrity is a great challenge.

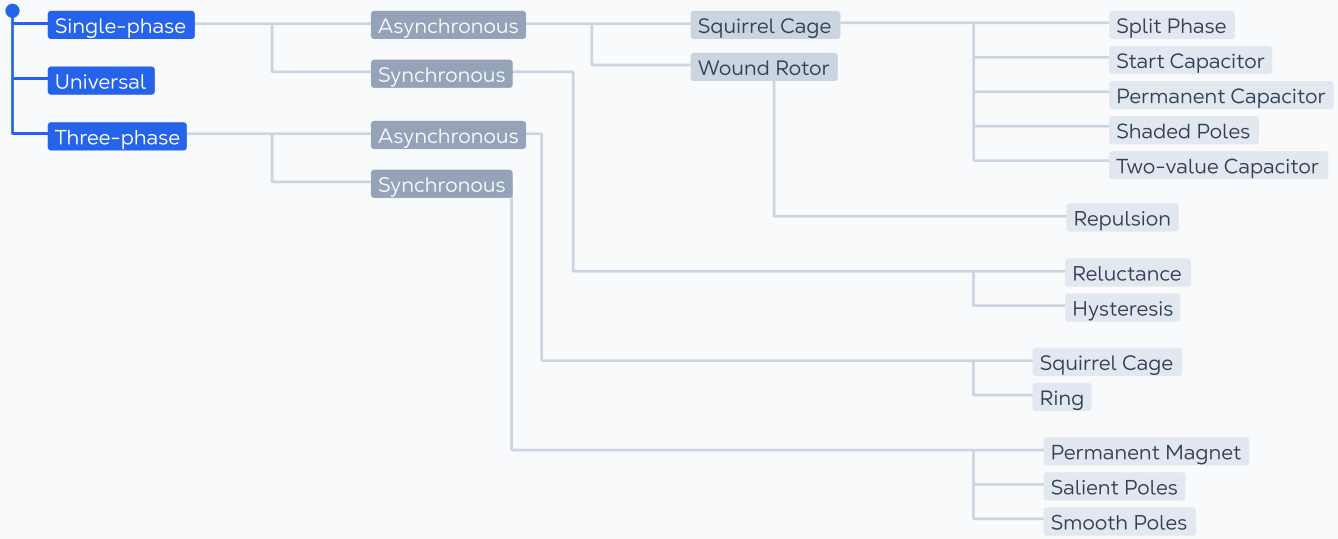


Types of Electric Motors

There are two main types of motors that branch into other subcategories:

ALTERNATING CURRENT MOTOR

Single-phase, Universal, and Three-phase.



DIRECT CURRENT MOTOR

Series excitation, independent excitation, compound excitation, and permanent magnet.



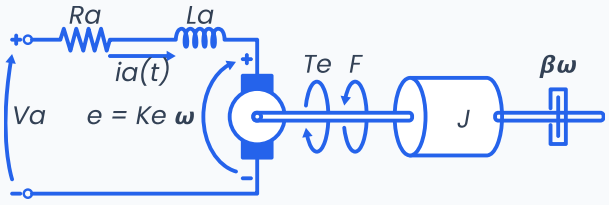
Direct Current Motors

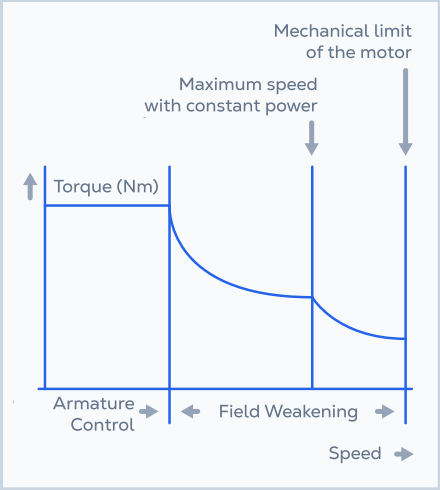
Direct current motors are used for applications that require variable speed, traction, and high starting torque. They are considered an important component for automation and control systems.

They are present in numerous automation processes, including rolling mills, extruders, presses, elevators, rolling mills, bobbin winders, and others.

The fundamental behavior of DC motors is where VaV_aVa (t) is the voltage applied by a source of continuous current. RaR_aRa is the armature resistance, LaL_aLa is the armature inductance, e(t)e(t)e(t) is the voltage induced in the armature, ω(t)\omega (t)ω(t) is the angular velocity, Te(t)T_e (t)Te(t) is the mechanical torque, FFF is the static friction, JJJ is the moment of inertia, and β\betaβ is the viscous friction.

ELECTROMECHANICAL DIAGRAM OF A DC MOTOR





CHARACTERISTICS

- High torque at the start and at low speeds
- Wide range of speed
- Ease of speed control
- High reliability
- Flexibility (various types of excitation)
- Relative simplicity of modern AC/DC converters

Single-Phase Alternating Current Motors

These are used when there is only one phase, mainly in applications that require lower power, generally less than 3kW.

It is important to highlight that they cannot start on their own, so it is necessary to use extra components such as models with start and permanent capacitors.

They are used in vacuum cleaners, fans, washing machines, refrigerators, some centrifugal pumps, etc.

Synchronous Motors

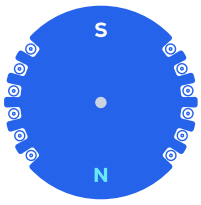
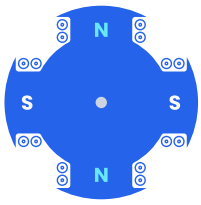
They are called Synchronous Motors because the rotor rotates at the same speed as the rotating magnetic field, produced in the three-phase winding of the stator. These are similar to alternators but differ in how the rotor winding is done.

The rotors are divided into two types: Salient Poles or Smooth Poles.

Synchronous motors have characteristics of high efficiency and torque, constant speed regardless of load variations, low maintenance cost, and power factor correction of the grid.

They are used in various areas:

TYPES OF POLES

SMOOTH POLES ROTOR	SALIENT POLES ROTOR
High Speed (2 to 4 poles)	Low Speed (4+ poles)
Steam or gas turbines	Hydraulic turbines
	
p = 1	p = 2

Mining
Conveyor Belts, Crushers, Mills, and others

Pulp & Paper
Extruders, Shredders, Defibrators

Sanitation
Pumps

Steel Industry
Rolling Mills, Fans, Pumps, and Compressors

Chemicals and Petrochemicals
Compressors, Mills, and others

Energy Transmission
Synchronous Compressors

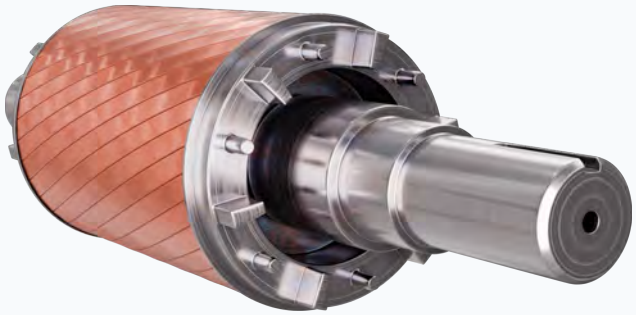
Rubber
Extruders, Mills, and Mixers

Cement
Conveyor Belts, Crushers, Mills, and others

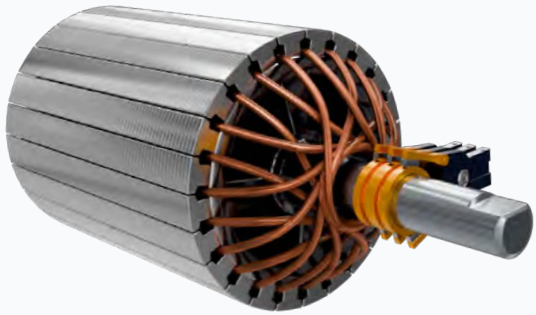
Three-Phase Induction Motors

They represent 90% of the power of manufactured motors. They are divided into two ctypes:

SQUIRREL CAGE ROTOR



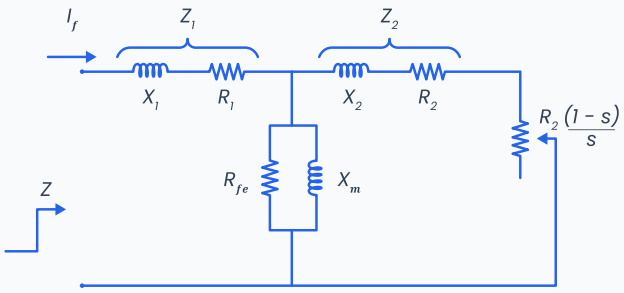
WOUND ROTOR



The three-phase induction motor has a winding responsible for the rotating magnetic field. Additionally, there can be a squirrel cage rotor that, when in contact with the rotating field, undergoes induction. It is called asynchronous because the speed of the rotating field is greater than that of the rotor. The difference between the two is called slip. The assembly of the winding is done according to the number of poles, slots, and power.

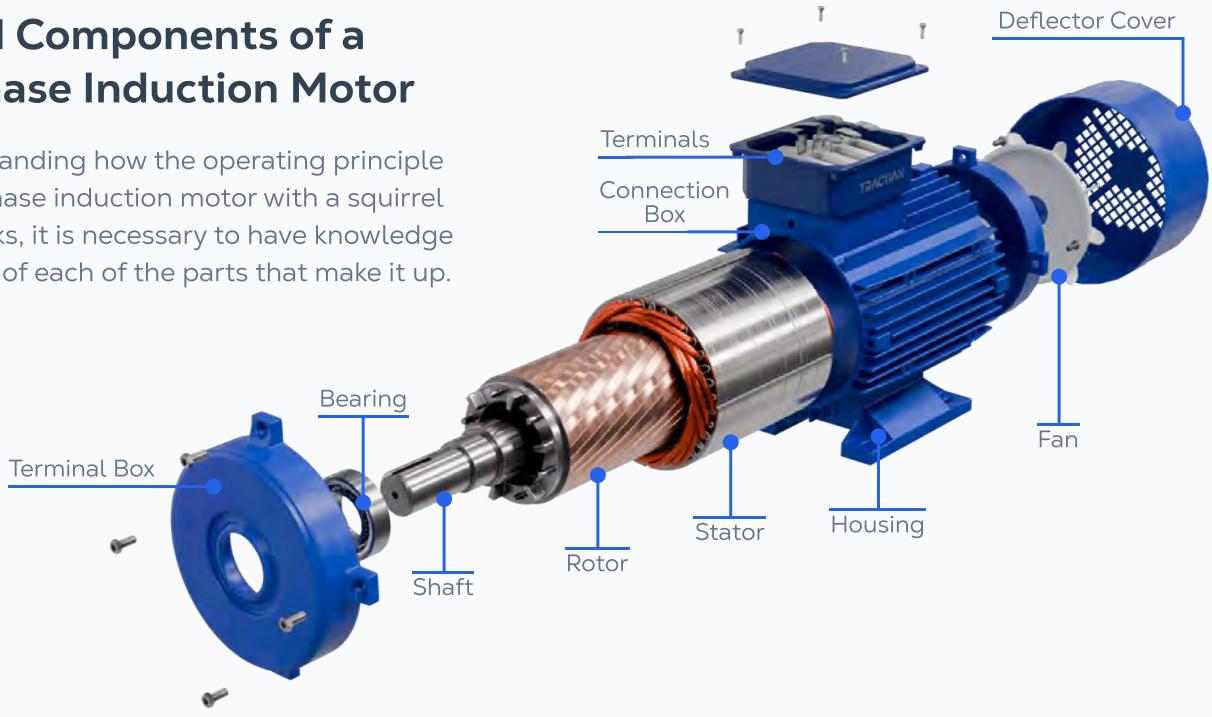
This type of motor is widely used due to its robust construction, reduced manufacturing cost, relatively low maintenance needs, and efficient operation.

EQUIVALENT CIRCUIT OF A THREE-PHASE MOTOR



Parts and Components of a Three-Phase Induction Motor

Before understanding how the operating principle of the three-phase induction motor with a squirrel cage rotor works, it is necessary to have knowledge of the function of each of the parts that make it up.



PARTS AND COMPONENTS

Stator

It is the part that remains fixed to the housing, with the purpose of conducting the magnetic flux to transform the kinetic energy of the armature.

The stator is made of laminated ferromagnetic material, usually silicon steel. The windings are housed in the slots of these laminations.

This lamination helps to increase insulation, and avoid eddy currents and parasitic losses, thereby increasing the machine's performance. In the stator, a magnetic field is formed capable of inducing a current in the rotor.



The core of the stator is made up of a stack of magnetic material sheets with low magnetic loss density (magnetic conductor).

The three-phase winding consists of three similar sets of coils – one for each phase – connected to the three-phase power supply.

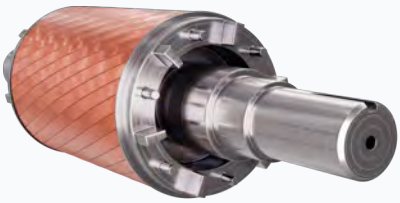
The housing is the structural component of the motor, housing, supporting, and protecting the static and rotating parts of the motor. It is made of a special aluminum alloy injected under pressure or gray cast iron, providing stiffness to the motor and low levels of vibration.

PARTS AND COMPONENTS

Rotor

The rotor is the rotating part of the motor. There are two types:

SQUIRREL CAGE ROTOR



Shaft • Core • Short-circuited rings • Conducting bars

WOUND ROTOR



Shaft • Core • Windings • Slip rings • Brushes

The shaft is the mechanical part where rotation occurs and the place where the rotor assembly is mounted, supported by bearings. It transmits the power developed by the motor and is heat-treated to avoid issues such as warping and fatigue.

The core is made of magnetic material, similar to the stator (magnetic conductor), with copper or aluminum bars placed parallel to each other and joined at the ends.

The cage has metal rings at the top and bottom, allowing current to circulate through them (electrical conductor).

Basically, the rotor is everything that rotates around its own axis, producing rotational movements.

End Cover

These are the unifying elements of the assembly that support the bearings.

These covers act as bearing housings, made from pressure-cast aluminum or cast iron. They are the components that provide the motor with the necessary high mechanical resistance.



Grounding

The motors have terminals for grounding located inside the junction box, made of brass to ensure perfect electrical contact.



Fan

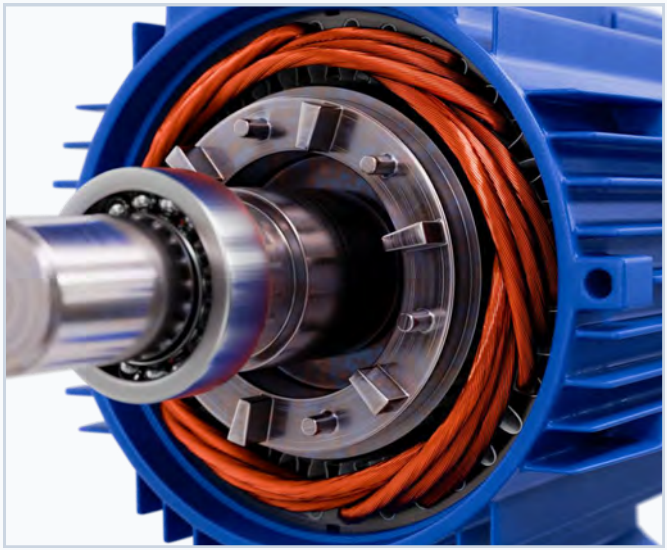
This part is designed to provide a forced ventilation system, preventing the motor from overheating.

The objective is to ensure maximum possible cooling, associated with reduced noise. The fan can be made of polypropylene or non-sparking aluminum.



Windings

All conductors used in the motor windings are made of copper, insulated with a polyester-based varnish. The stator insulators can be of some insulation classes depending on the operating temperature of the motor.



Junction Box

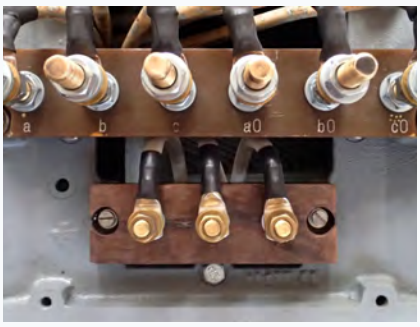
Its main function is to house the power cables, reinforcing the external environment and protecting the cables from any possible accidental physical contact.

Therefore, the junction box can be made of steel sheet or pressure-injected aluminum, allowing a 90° rotation to facilitate assembly.



Terminal Plate

The terminal plates are made of self-extinguishing, non-hygroscopic material, resistant to current leaks and with high dielectric rigidity.



Key

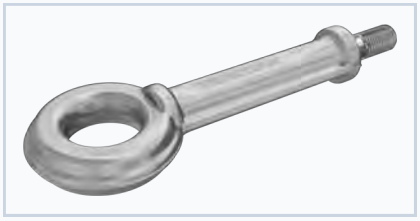
It is designed to ensure the perfect fixation of the load element hub to the motor shaft. It is made of SAE 1045 steel.



Lifting Eye

It is designed to facilitate the handling, transportation, and installation.

Its installation is done on motors according to the type of housing, and it can be fixed or made of forged steel, threaded into the housing itself.

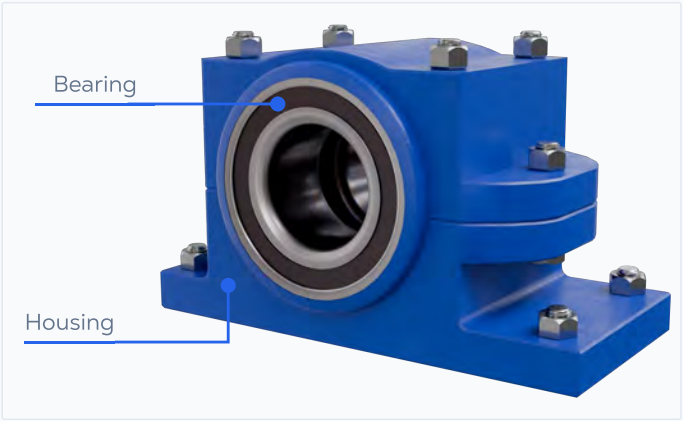


Bearing

This is a mechanical element of fundamental importance for motors, both in the construction phase and during operation.

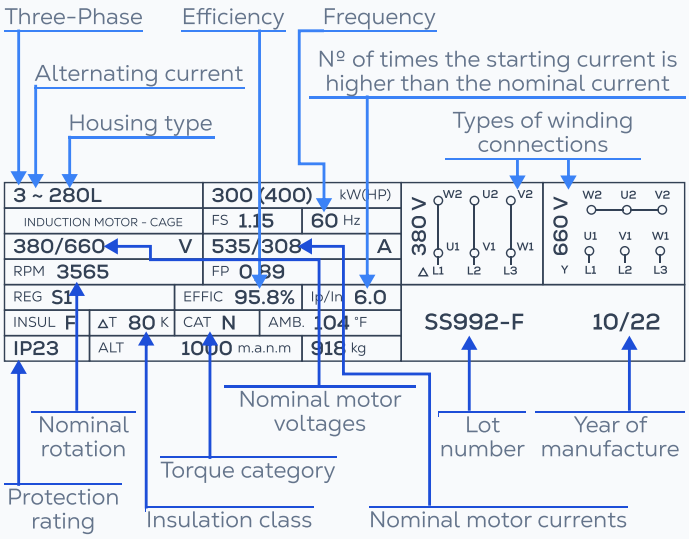
The bearing supports the internal moving elements of the motor and assists in the transmission of motion with minimal friction. Therefore, it is essential to maintain the predetermined clearance limits between the stator and rotor based on the air gap.

Rolling bearings can be lubricated with grease, while sliding bearings are lubricated with oil.



Nameplate

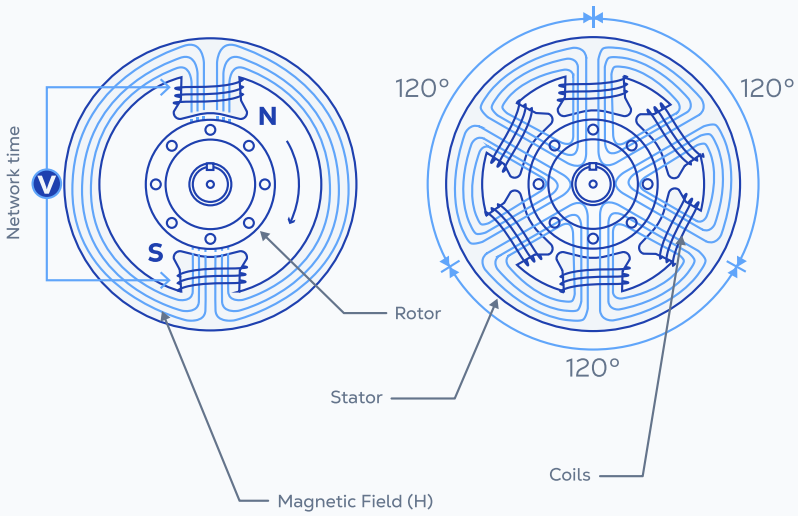
It is the element used for gathering the main information for proper motor installation and maintenance.



Operating Principles

The operating principle of these machines is to use the phenomenon of electromagnetic induction in conductors to manipulate magnetic fields and thus generate rotor movement. With current passing through the windings, a magnetic flux is generated that travels through the stator core.

When three pairs of coils are powered by a three-phase network, the fluxes interact to form a rotating field that rotates around the equipment.





The rotor consists of a core of steel laminations with short-circuited conductors arranged parallel to each other and embedded in slots around the core perimeter.

As the current passes through the rotating magnetic field in a conductor, it induces a current that travels the length of the rotor by the Lorentz force, inducing forces that form the field and the current, generating the rotational movement of the entire system.

Why Do Motors Burn Out?

The motor depends on the condition of the windings and adequate power supply for electromagnetic induction to occur properly and generate rotation. Burnout occurs when some operational condition induces a high current that heats and burns the c onductor, preventing its operation.

Short Circuit

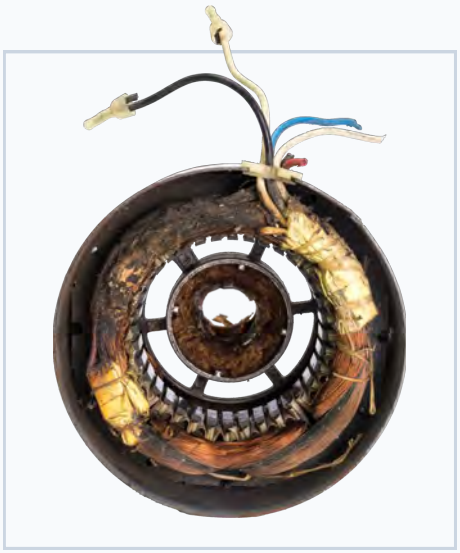
The copper conductors are coated with an insulating enamel layer, making it possible to wind the wires of the windings one on top of the other without an electrical short circuit.

Additionally, the enameled conductors are joined together with a varnish to prevent them from vibrating against each other, which could damage the enamel.

The windings are also wrapped in an impregnating resin, which fills the empty spaces between the wire slots in the stator, aiding in the dissipation of the heat generated by the conductor.

This entire winding insulation system exists to prevent electrical contact between the conductors, which could lead to a short circuit and consequent conductor burnout. These short circuits can occur between turns of the same winding, between different windings, or even in the motor connection terminals.

Insulation defects are associated with some contamination inside the motor, such as the degradation of the insulating material due to excessive temperature, or from abrasion and rapid oscillations in the power supply voltage.





Overheating

The passage of electrical currents in the windings generates heating of the components due to the Joule effect. Therefore, the generated heat needs to be released so as not to cause overheating and thus deteriorate the insulation of the conductors.

If the temperature of the windings is above the insulation limit, gradual and generalized aging will occur, reducing the useful lifespan of the asset. It is important to remember that all motors have a ventilation system attached to the rotor, which helps to remove internal heat. Any obstruction of airflow can cause overheating; the most common causes are the accumulation of dirt, dust, grease, and rust.

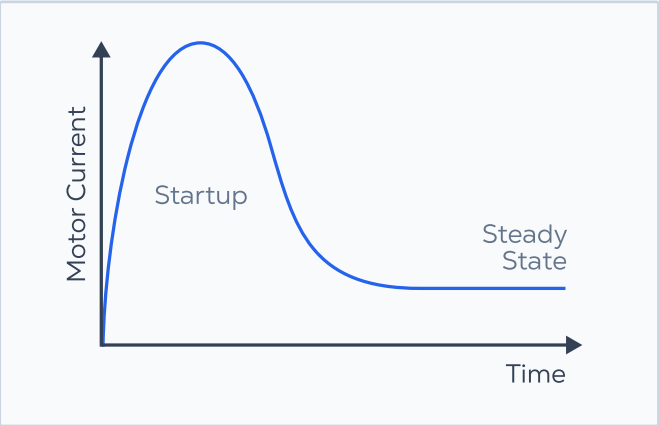


Temperature Increase ΔΘ (°F)	0	3.6	7.2	10.8	14.4	18.0	21.6	25.2	27.0	28.8	32.4	36.0
Useful Life Years	20	18.46	17.04	15.73	14.54	13.46	11.49	10.62	9.84	9.15	6.91	4.31
Useful Life Reduction %	0	7.7	14.8	21.4	27.3	32.7	42.5	46.9	50.8	54.3	65.4	78.4

Locked Rotor

This condition imposes an extreme overload on the induction motor. The high current of a locked rotor causes all the input energy to be converted into heating of the rotor and stator.

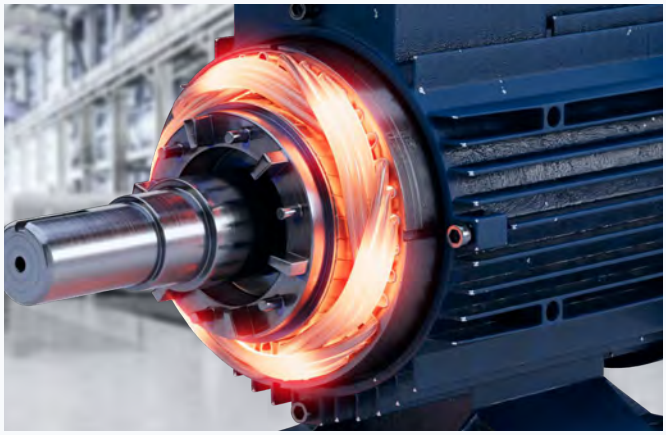
At startup with 100% load, the currents in the stator windings vary from 3 to 7 times, and if the motor cannot overcome its startup due to the lock, the current will remain at the startup level causing overheating.



Voltage Spikes

These failures occur due to an increase in voltage in one or more phases, which means that if the motor is not protected, it will burn out.

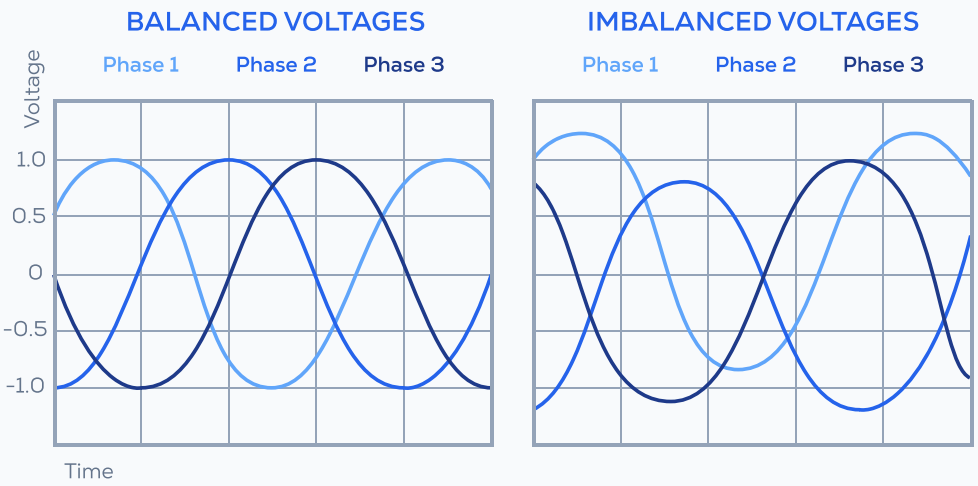
This cause can be related to lightning strikes, where energy travels through the electrical network, oscillating the power supply violently. Additionally, they can be caused by maneuvers in capacitor banks or problems in the motor drive and control system.



Voltage Imbalance

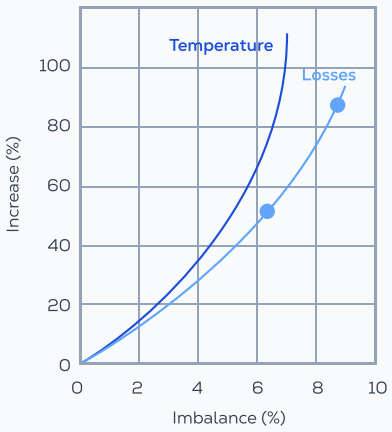
It is a condition where the phases have different voltage amplitudes or an angular phase shift different from 120°, or both conditions simultaneously.

Voltage imbalance can cause energy waste due to the existence of greater losses caused by high unbalanced currents.



Voltage imbalances of 3.5% can increase motor losses by up to 20%, and imbalances above 6.5% cause immediate operational problems. Small imbalances of 1% to 2% are also harmful as they cause an increase in energy consumption, and if not detected, the motor will be systematically overloaded. Therefore, the network voltage should be frequently monitored, and any imbalance greater than 1% should be corrected.

Voltage imbalance causes operational effects on the motor, such as the emergence of torque in the opposite direction to hinder movement, changes in start time, and reduction of the power factor.

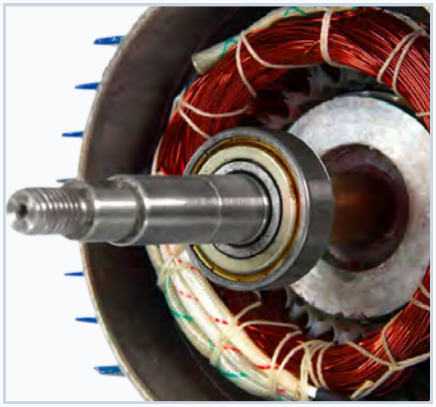


Phase Failure

This condition occurs when there is a problem with the power supply, resulting from a broken power cable, a blown fuse, a failed transformer phase, or issues in the drive system.

If the motor is not in operation and a phase failure occurs, it will not start. But if it is already running, the torque will be significantly reduced and may continue at low speed.

Phase failure is the extreme condition of phase imbalance, as a considerable amount of current flows through the remaining phases, causing motor overheating.

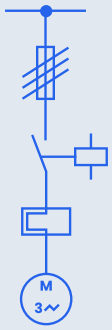


Protection Components

Motors are susceptible to a series of phenomena that can lead to equipment burnout. Therefore, we must use resources that will protect the equipment in case of overheating to prevent damage to the windings. The fuse is the most basic element and protects against high current spikes but does not perform well against other conditions.

Thermal protection using an overload relay monitors the temperature through the current passage and isolates the system if necessary. With thermal protection using probes, the protection becomes even more precise and reduces the reaction time directly in the equipment.

By combining these components, a protection circuit is created that avoids burnout and prolongs the equipment's useful lifespan.

Causes of Overheating	Protection based on current		
	Fuse 	Fuse and thermal protector 	Protection with thermal probes in the motor 
Overload with current 1.2 times the nominal current	●	●	●
Locked Rotor	◐	●	●
Phase Failure	●	◑	●
Excessive Voltage Variation	●	◑	●

● Not protected ◐ Partially protected ● Totally protected

Common Failures

You know that mechanical or electrical problems can cause electric motors to fail, leading to production line downtime and significant losses for any company.

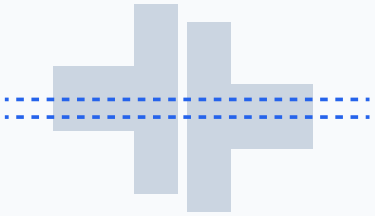
The issues that lead to failures and their components can be of various natures. However, most can be avoided with well-trained teams and the application of effective maintenance programs, as well as condition monitoring systems capable of predicting potential problems.

Next, we list the most common failures that occur in this machine category.

Misalignment

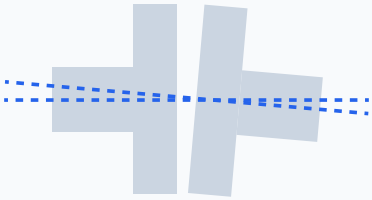
This problem occurs when the drive shaft of the motor (rotor) or the coupling piece is not correctly aligned with the load, resulting in the transfer of mechanical stresses that increase wear on the motor and its apparent mechanical load. One of the effects of this problem is increased vibration, both on the load and the motor itself. Misalignment can be classified into three types: parallel, angular, or combined.

PARALLEL OR RADIAL MISALIGNMENT



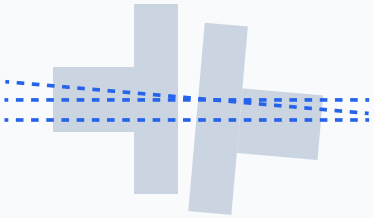
Occurs when the center lines are parallel.

ANGULAR OR AXIAL MISALIGNMENT



Is observed from an angle formed between the center lines when they cross.

COMBINED MISALIGNMENT



Involves both parallel and angular deviations in the center lines, and is considered the most common of the three types.

Overload

This occurs when a motor is required to operate beyond its nominal torque, meaning the operating electrical current is above normal, causing overheating. This decreases the motor's useful life, and depending on the level of overload, the protection device of the motor may trip, causing unexpected operational downtime.

When a motor is overloaded, it is possible to observe some signs, such as excessive consumption of electrical current or insufficient torque. The excessive heat caused by the overload is one of the main causes of failure and the primary damage is the deterioration of the winding insulation, which can lead to permanent failure.

Additionally, excess material, starts, inertia, reverse operations, improper operation, vibration, and time also contribute to the increase in temperature.



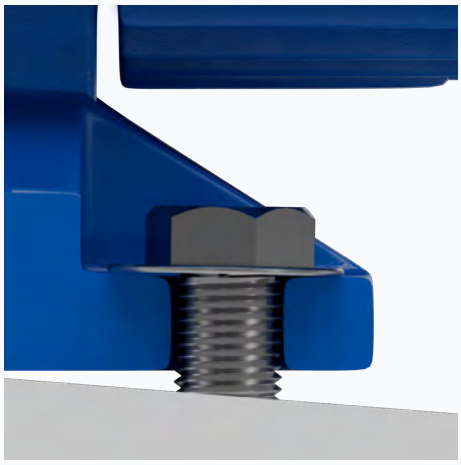
Soft Foot

This occurs when the mounting feet of the motor or its drive component are not seated on the same surface, causing mechanical misalignment stresses on both the motor shafts and the load.

This failure can create new mechanical misalignment stresses. One way to detect the problem is by observing the mounting bolts: when soft foot occurs, the bolts are usually in diagonal positions.

A simple analogy to visualize this condition is to think of sitting on a wobbly chair: it will rock diagonally in its unsteady position.

To avoid this issue, it is important that the motor and load are firmly secured to the surface.



Bearing Wear

It is estimated that a large part of motor failures are caused by bearing wear.

Bearing failures can occur for various reasons:

BEARING FAILURES: MAIN CAUSES

Inadequate Lubrication	34%
Misalignment	20%
Contamination	19%
Assembly Errors	18%
Unknown Causes	5.5%
Inadequate Storage and Handling	2.8%

In general terms, we can classify bearing failures into pre-operational and operational.

PRE-OPERATIONAL

Before or during bearing installation

- Static misalignment
- Imprecise adjustment of the shaft and bearing housing
- Bearing seats on the shafts and defective bearing housings
- Handling and transportation
- Storage
- Excessive voltage due to electrical current passing through the bearing

OPERATIONAL

During asset operation

- Inefficient lubrication or sealing
- False brinelling
- Material fatigue
- Current leak (due to electrical current passing through the bearing)
- Operational misalignment

COMMON FAILURES

Shaft Unbalance

This occurs when a rotating part of the center of a mass does not rotate over the axis of rotation, causing unbalance. The irregular distribution of mass generates centrifugal forces that are transmitted to the bearings and can damage components, resulting in reduced asset lifespan.

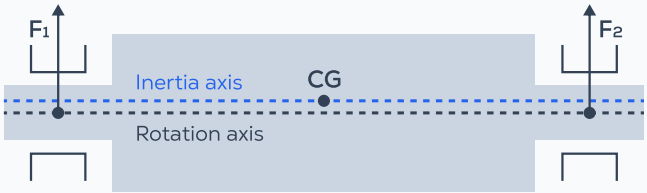
Unbalance can be caused by dirt buildup, lack of balancing weights, irregular mass in the motor windings, or even manufacturing variations.

The main damage is premature wear of mechanical transmission components, which consequently results in premature asset failures.

Unbalance can also be classified into three types: static, coupled, and dynamic.

STATIC UNBALANCE

The principal axis and the rotation axis around which the machine rotates appear parallel, but the masses do not coincide.



COUPLED UNBALANCE

The axes are not parallel, but have an intersection between the mass center of gravity and the rotation axis.



DYNAMIC UNBALANCE

It is a combination of static and coupled unbalance, where the principal axis and the rotation axis do not align and are not parallel.



Mechanical Looseness

Looseness occurs when there is excessive wear between parts of an asset. It can happen due to wear, excessive clearance between fixed and rotating elements, or looseness in parts that are normally stationary, such as a base or foundation.

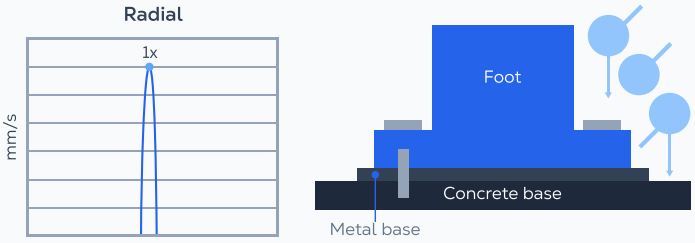
There is a third definition of looseness according to the Technical Associates of Charlotte Vibration Diagnostic Manual, and it is classified as types A, B, and C. We will analyze them next.

Factors that Intensify Mechanical Looseness

- Erosion or sinking of the ground under the equipment
- Loose base screws
- Damage to the machine base structure

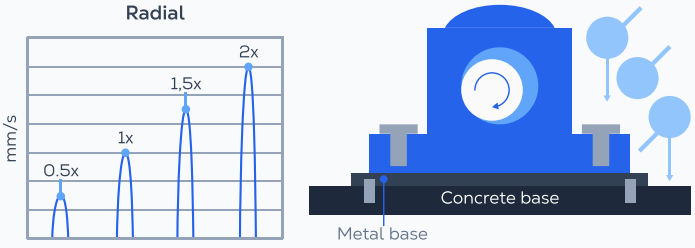
MECHANICAL LOOSENESS TYPE A

Type A looseness, or structural looseness, is caused by fragility at the base, or foundation of the asset, and generally occurs due to frame distortions, deterioration of the base joint, or looseness of the screws that support the base. It manifests in the spectrum as a wave of one pulse per rotation.



MECHANICAL LOOSENESS TYPE B

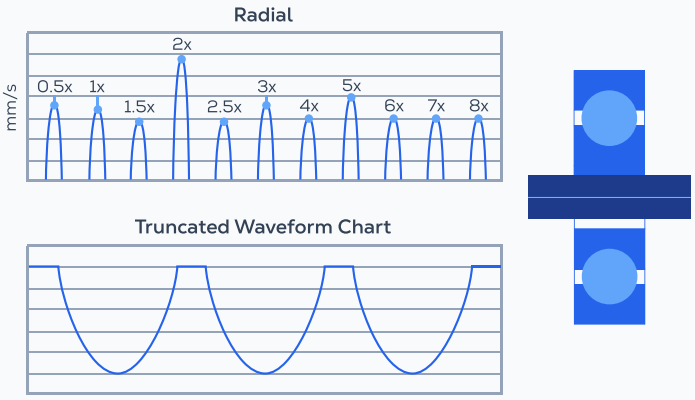
Type B looseness is characterized by the presence of loose screws on the base support, cracks in the bearing pedestal and/or in the skid structure. In the spectrum, it appears as a temporal wave of two pulses per rotation.



MECHANICAL LOOSENESS TYPE C

Type C looseness occurs due to incorrect adjustment between the parts that make up the rotor's dynamic forces, such as excessive clearance between the bearing and sleeve, a loose rotor on a certain axis, or a loose bushing on the cover, for example.

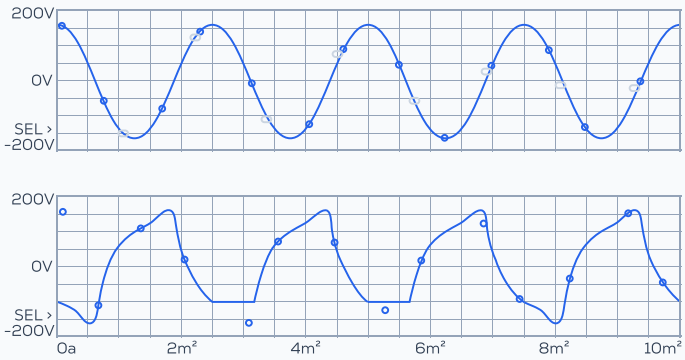
In the spectrum, it presents a high noise floor, detecting the looseness, and several harmonics, which represent the nonlinear responses of the loose parts.



COMMON FAILURES

Harmonic Distortion

Harmonics are high-frequency components of an electrical signal. Basically, they are any unwanted high-frequency waves that provide unwanted energy to the windings of the motor.



Although this additional energy is not used to drive the rotor of the motor, it circulates in the bearings, contributing to internal energy losses (which occur in the form of heat generation).

Over time, these losses can lead to the deterioration of the winding insulation capacity. Consequently, this can cause problems such as an increase in the operating temperature of the asset, a gradual reduction in the motor's efficiency, and additional maintenance costs.

Any system that serves electronic loads is subject to harmonic distortion. However, it is necessary to investigate the sources of these distortions to analyze if these occurrences are within normal limits and can be avoided without harming your asset.

COMMON FAILURES

Sigma Current

These are parasitic currents that circulate through an electrical circuit, generated by frequency in the signal, voltage level, capacitances, and associated inductances of the electrical conductors. By circumventing the grounding protection system, sigma currents can cause unintentional trips and even excessive heating of the motor windings.

Sigma currents are also found in motor cables, representing a sum of various frequencies that are not eliminated over time. A simple way to identify that a unit has sigma currents is to always ensure that there is a grounding return cable from the unit to the generator, or that the current leaves the unit.

To prevent this issue, it is recommended that electrical connections are dimensioned and of high quality. It is also important to verify that there are no improper connections or unbalanced loads on the conductor.

How to Avoid These Conditions?

Preventive methods combined with predictive maintenance (PdM) always emerge as the best alternative, as they are able to detect failures and assist in maintaining proper asset operation and reliability.

Techniques such as motor circuit analysis, thermography, vibration monitoring, and energy monitoring help maintain assets and improve maintenance routines, thereby extending asset life cycles.

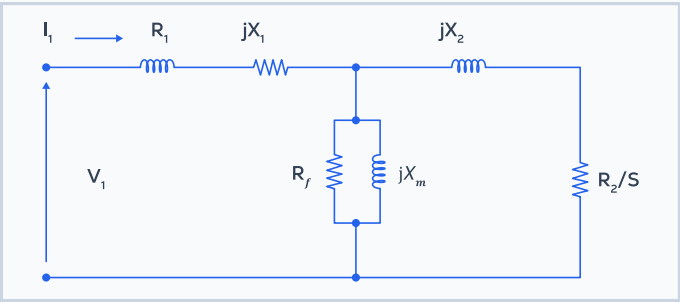
Motor Circuit Monitoring | Thermography | Vibration Monitoring | Energy Monitoring

HOW TO AVOID THESE CONDITIONS?

Motor Circuit Analysis

This is a technique used since 1985 that employs a series of low-voltage tests to detect and monitor faults in motors and/or unbalanced stator windings. Also known as MCA (Motor Circuit Analysis), it allows for the identification of failures at the beginning of the asset's useful lifespan, such as unbalanced voltages or degradation.

Martínez (2008) points out that the test considers that most motors have a homogeneous distribution of 120° between phases and that the phases of the power supply source also have a homogeneous 120° phase shift. According to the same author, in each phase when the voltage increases, the current increases proportionally in each phase and is limited by the impedance of the motor circuit.



The tests that comprise the technique:

- **Resistance measurement**
- **Impedance measurement**
- **Inductance measurement**
- **Phase angle measurement**
- **Current/frequency response measurement**
- **Insulation resistance measurement**

Thermography

One of the biggest indicators of electrical failures or overloads in factories and industries is the excessive increase in temperature.

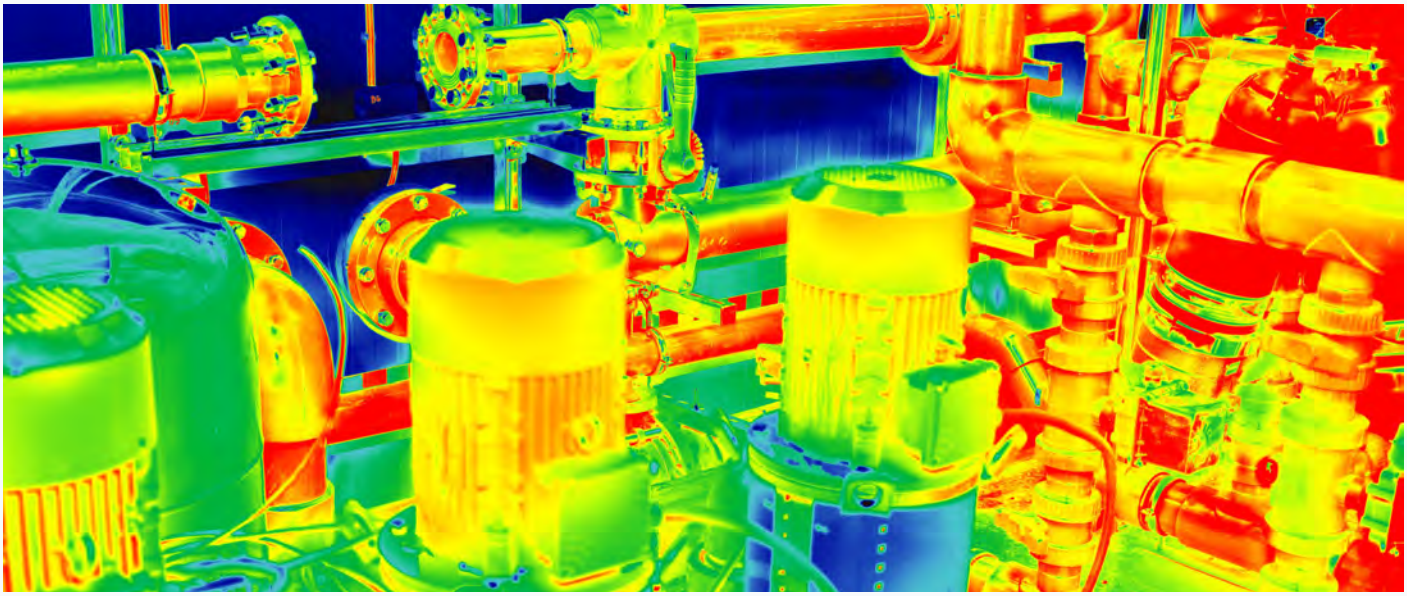
Thermography maps heat and radiation patterns in machines, indicating faults or degradations. The higher the temperature, the greater the infrared radiation emitted by the object.

It extends human vision through the infrared spectrum—a frequency of electromagnetic radiation emitted by any body, with intensity proportional to its temperature.

The colorimetric scale (temperature scale) goes from black (low temperature) to white (highest temperature) through several variations of warm colors, which depend on the scale used. In thermography, the most common scale is called IRON, which goes from black to white through shades of blue, green, yellow, red, orange, and white.

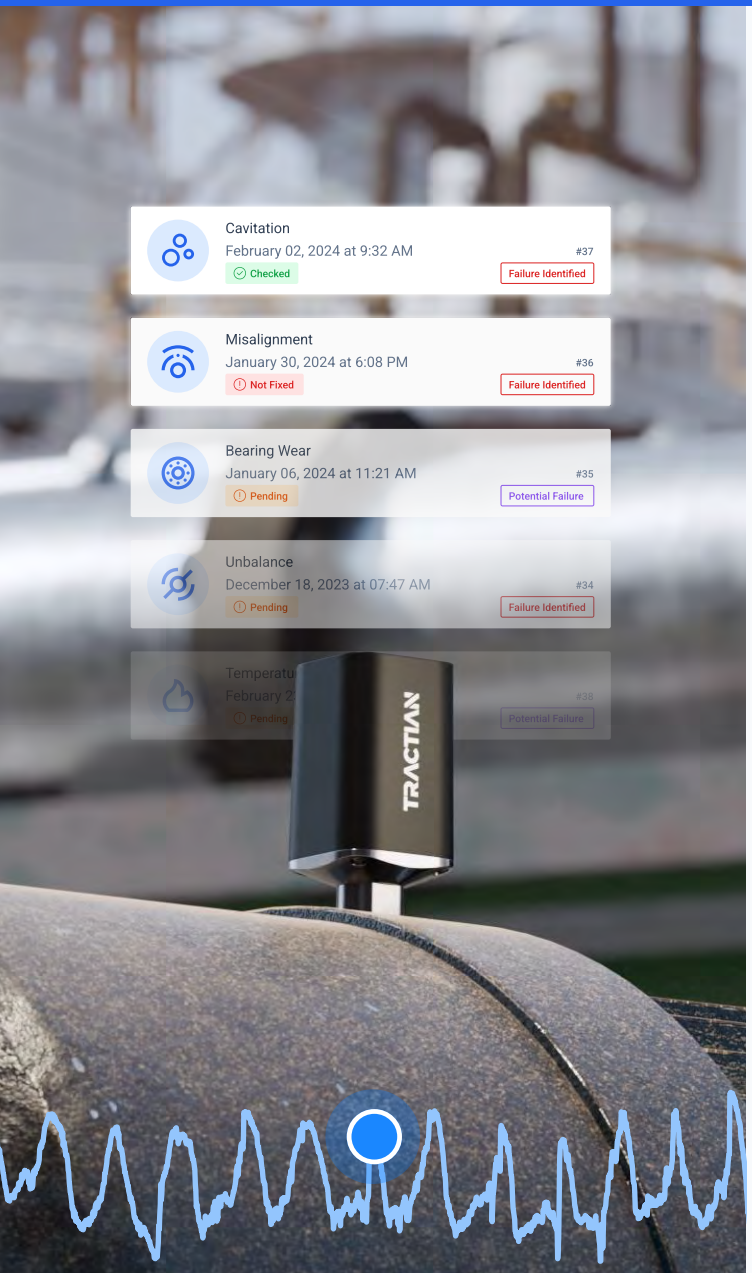
In electric motors, this method is important for identifying thermal faults and excessive bearing wear, without needing to disassemble the motor.

Reminder: Each motor is designed to operate at a specific internal temperature.





Color	Wave length (nm)	Frequency (THz)
Red	~ 625-740 nm	~ 480-405 THz
Orange	~ 590-625 nm	~ 510-480 THz
Yellow	~ 565-590 nm	~ 530-510 THz
Green	~ 500-565 nm	~ 600-530 THz
Cyan	~ 485-500 nm	~ 620-600 THz
Blue	~ 440-485 nm	~ 680-620 THz
Violet	~ 380-440 nm	~ 790-680 THz



HOW TO AVOID THESE CONDITIONS?

Vibration Monitoring

By monitoring vibration in electric motors, it is possible to identify and anticipate failures such as misalignment, unbalance, mechanical looseness, bearing wear, lubrication failures, overload, and many others.

Smart Trac Ultra, an IoT vibration sensor powered by Auto Diagnosis™, monitors not only motors, but over 100 machine categories including pumps, gearboxes, CNCs, mills, and bearings.

Using AI-powered fault-finding capabilities, it provides a comprehensive assessment of the asset's health and operating behavior, detects potential failures before they compromise production lines, and provides detailed diagnostics so maintenance and reliability teams can act on the issues before they escalate.

Learn more about how Smart Trac Ultra can improve machine availability at your facility.

Click Here

HOW TO AVOID THESE CONDITIONS?

Monitoring Electrical Energy

By monitoring power sources, it is possible to identify failures in the electrical network that can damage the electric motor over time, such as overvoltage, undervoltage, low power factor, and phase imbalance.

It is possible to act on potential problems in the electrical network, make better load distributions, resize projects and install capacitor banks to improve the energy quality delivered to the assets.

It is not enough to inspect these conditions periodically; it is essential to continuously monitor them to ensure good operation and long component lifespans while increasing asset efficiency.

Energy Trac samples voltage, current, power, and consumption data from industrial assets automatically and in real time, alerting teams to failures in the network.

Some of the detected failures are: short circuit in the windings, overload, overcurrent, phase failure, imbalance in currents and voltages, insulation failure between phases or phases to ground.

It is also a great tool in understanding consumption patterns for different assets within an industrial plant and increasing energy efficiency, generating significant cost savings.

Explore how Energy Trac can boost energy efficiency at your company and give your team insights on energy consumption.

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TRACTIAN

tractian.com

get@tractian.com

201 17th Street NW, 2nd Floor, Atlanta - GA, 30363

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